

Tic Tac Toe Game Program

This project develops a simple Tic Tac Toe game where two players take turns placing X and O symbols on a 3x3 grid. The objective is to create a horizontal, vertical, or diagonal line of three symbols to win the game. A reset option is also included to start a new game.

Python Program (Tkinter GUI)

```
from tkinter import *
from tkinter import messagebox

root = Tk()
root.title("Tic Tac Toe")

current_player = "X"
board = [""] * 9

buttons = []

def check_winner():
    winning_positions = [
        (0,1,2), (3,4,5), (6,7,8),
        (0,3,6), (1,4,7), (2,5,8),
        (0,4,8), (2,4,6)
    ]

    for a, b, c in winning_positions:
        if board[a] == board[b] == board[c] != "":
            return board[a]

    if "" not in board:
        return "Tie"

    return None

def button_click(index):
    global current_player

    if board[index] == "":
        board[index] = current_player
        buttons[index]["text"] = current_player

        winner = check_winner()

        if winner:
            if winner == "Tie":
                messagebox.showinfo("Game Over", "It's a Tie!")
            else:
                messagebox.showinfo("Game Over", f"Player {winner} wins!")
                reset_game()
        else:
            current_player = "O" if current_player == "X" else "X"

def reset_game():
    global current_player, board

    current_player = "X"
    board = [""] * 9

    for button in buttons:
        button["text"] = ""

for i in range(9):
    button = Button(root, text="", width=10, height=4,
                    font=("Arial", 20),
                    command=lambda i=i: button_click(i))
    button.grid(row=i//3, column=i%3)
    buttons.append(button)

reset_button = Button(root, text="Reset Game",
                      font=("Arial", 14),
```

```
        command=reset_game)
reset_button.grid(row=3, column=0, columnspan=3, sticky="we")

root.mainloop()
```

Feature	Description
Two Player Mode	Players alternate between X and O
Winner Detection	Checks rows, columns, and diagonals
Tie Detection	Declares tie if grid is full
Reset Option	Starts a new game instantly

This Tic Tac Toe project is beginner-friendly and demonstrates GUI development, game logic, event handling, and conditional statements using Python Tkinter.